



NCI Protocol #: 10323

Patient ID:

Date:

1 of 16

Patient Name:

Date Collected:

Biopsy Site: **Lymph Node**

Sample ID:

Telephone:

Tumor Content (%): **70%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Metastatic melanoma**

MoCha ID:

Fax:

Tumor content assessment performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Melanoma Findings

Gene	Finding
BRAF	BRAF V600E
KIT	Not detected
NRAS	Not detected
NTRK1	Not detected
NTRK2	Not detected
NTRK3	Not detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	BRAF V600E	atezolizumab + cobimetinib + vemurafenib ¹ binimetinib + encorafenib ¹ cetuximab + encorafenib ¹ cobimetinib + vemurafenib ¹ dabrafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ vemurafenib ¹ dabrafenib + pembrolizumab + trametinib	binimetinib + encorafenib ¹ cetuximab + encorafenib ¹ dabrafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ encorafenib + panitumumab vemurafenib	53

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
III	PDGFRA P895S	None	imatinib	4
IIC	ARID1A Q766*	None	None	6
IIC	IDH2 R172K	None	None	5

Public data sources included in relevant therapies: FDA¹, NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants							
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
ARID1A	p.(Q766*)	c.2296C>T	.	chr1:27088687	38.75%	NM_006015.4	nonsense
PDGFRA	p.(P895S)	c.2683C>T	.	chr4:55154974	42.54%	NM_006206.4	missense
BRAF	p.(V600E)	c.1799T>A	COSM476	chr7:140453136	57.01%	NM_004333.4	missense
IDH2	p.(R172K)	c.515G>A	COSM33733	chr15:90631838	36.82%	NM_002168.2	missense

Comments:

N/A

Signed By: _____

Biomarker Descriptions

ARID1A (AT-rich interaction domain 1A)

Background: The ARID1A gene encodes the AT-rich interaction domain 1A tumor suppressor protein¹. ARID1A, also known as BAF250A, belongs to the ARID1 subfamily that also includes AR1D1B^{1,2}. ARID1A and ARID1B are mutually exclusive subunits of the BAF variant of the SWI/SNF chromatin-remodeling complex^{2,3}. The BAF complex is a multisubunit protein that consists of SMARCB1/IN1, SMARCC1/BAF155, SMARCC2/BAF170, SMARCA4/BRG1, and SMARCA2/BRM³. The BAF complex remodels chromatin at promoter and enhancer elements to alter and regulate gene expression^{3,4}. ARID1A binds to transcription factors and coactivator/corepressor complexes to alter

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.05(003).

Biomarker Descriptions (continued)

transcription². Recurrent inactivating mutations in BAF complex subunits, including ARID1A, lead to transcriptional dysfunction thereby, altering its tumor suppressor function².

Alterations and prevalence: Mutations in SWI/SNF complex subunits are the most commonly mutated chromatin modulators in cancer and have been observed in 20% of all tumors⁴. The majority of ARID1A inactivating mutations are nonsense or frameshift mutations². Somatic mutations in ARID1A have been identified in 50% of ovarian clear cell carcinoma, 30% of endometrioid carcinoma, and 24-43% of uterine corpus endometrial carcinoma, bladder urothelial carcinoma, and stomach adenocarcinoma^{3,5,6}. In microsatellite stable (MSS) colorectal cancer, mutations in ARID1A have been observed to correlate with increased tumor mutational burden (TMB) and expression of genes involved in the immune response⁷.

Potential relevance: Currently, no therapies are approved for ARID1A aberrations.

BRAF (B-Raf proto-oncogene, serine/threonine kinase)

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival^{8,9}. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF¹⁰. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead^{10,11,12}.

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)^{5,6,13,14,15}. Mutations at V600 belong to class 1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types^{11,16}. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions¹¹. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and N581S/I¹¹. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms^{17,18,19}. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in exon 11 as well as additional codons flanking V600 in the activation loop¹⁶. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer^{5,6}. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types^{20,21,22,23,24}. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation^{12,20,22}.

Potential relevance: Vemurafenib²⁵ (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive¹¹. BRAF kinase inhibitors including dabrafenib²⁶ (2013) and encorafenib²⁷ (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib²⁷ is approved in combination with cetuximab (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations¹¹. Trametinib²⁸ (2013) and binimetinib²⁹ (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib³⁰ (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The pan-RAF kinase inhibitor DAY-101 was granted breakthrough therapy designation (2020) by the FDA for pediatric patients with advanced low-grade glioma harboring activating RAF alterations³¹. The ERK inhibitor ulixertinib³² was also granted a fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma³³. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)^{34,35,36,37,38,39,40}. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported²⁴.

Biomarker Descriptions (continued)

IDH2 (isocitrate dehydrogenase (NADP(+)) 2)

Background: The IDH1 and IDH2 genes encode homologous isocitrate dehydrogenase enzymes that catalyze the conversion of isocitrate to α -ketoglutarate (α -KG)⁴¹. The IDH1 gene encodes the NADP+ dependent cytoplasmic isocitrate dehydrogenase enzyme; IDH2 encodes the mitochondrial isoform.

Alterations and prevalence: Recurrent somatic mutations in IDH1 and IDH2 are mutually exclusive and observed in several malignancies including glioma, chondrosarcoma, intrahepatic cholangiocarcinoma, acute myeloid leukemia (AML), and myelodysplastic syndrome (MDS)⁴². Recurrent IDH2 variants include predominately R140Q and R172K plus other substitutions at lower frequencies. These gain of function variants confer neomorphic enzyme activity⁴³. Although wild-type enzymatic activity is ablated, recurrent IDH2 variants catalyze the conversion of α -KG to D-2-hydroxyglutarate, an oncometabolite with diverse effects on cellular metabolism, epigenetic regulation, redox states, and DNA repair^{41,44}. Recurrent IDH2 mutations are present in 10-20% of patients with AML and 5% of patients with MDS^{45,46,47}.

Potential relevance: Enasidenib⁴⁸ is FDA approved (2017) for the treatment of AML patients with IDH2 R140G/L/Q/W and R172G/K/M/S/W mutations. In AML, acquired resistance to enasidenib has been associated with the emergence of Q316E or I319M mutations⁴⁹. IDH2 R172 and R140Q variants are associated with poor prognosis in MDS but have been shown to confer improved prognosis in lower grade gliomas^{50,51,52}. Additionally, IDH2 mutations are associated with inferior overall survival in polycythemia vera (PV) and essential thrombocythemia (ET) as well as inferior leukemia-free survival in primary myelofibrosis (PMF)⁵³.

PDGFRA (platelet derived growth factor receptor alpha)

Background: The PDGFRA gene encodes the platelet derived growth factor receptor alpha, a member of the PDGF receptor type III receptor tyrosine kinase family, which includes PDGFRB, CSF1R, FLT1, FLT3, FLT4, KDR, and KIT^{54,55}. PDGFRA is a receptor for platelet derived growth factors, which are mitogens for cells of mesenchymal origin⁵⁶. PDGFRA may function as a homodimer or heterodimer with PDGFRB depending on the ligand⁵⁷. The PDGFRA gene is physically adjacent to KIT and KDR on chromosome 4q12. Ligand binding to PDGFRA results in kinase activation and stimulation of downstream pathways including the RAS/RAF/MEK/ERK and PI3K/AKT/MTOR pathways promoting cell proliferation and survival.

Alterations and prevalence: Recurrent somatic PDGFRA alterations are observed in both solid and hematological cancers and include activating mutations, gene amplification, and translocations generating PDGFRA gene fusions. Recurrent PDGFRA activating mutations, including D842V, V561D, N659K, and in-frame deletions in exon 18, are common in 30-40% of KIT negative gastrointestinal stromal tumors (GISTs) and approximately 7% overall^{58,59,60,61}. PDGFRA recurrent mutations are also described in adult and pediatric glioblastoma and high-grade gliomas^{61,62}. In these cases, PDGFRA amplification is common (about 10% of cases) and recurrent mutations frequently co-occur with gene amplification^{5,6}. PDGFRA fusions are observed in gliomas and glioblastomas as well as eosinophilic leukemias, of which the FIP1L1-PDGFRB fusion defines approximately half of patients with hypereosinophilic syndrome^{63,64,65}.

Potential relevance: The FDA has granted fast track designation to crenolanib⁶⁶ (2017) for GISTs harboring PDGFRA D842V mutation. Avapritinib⁶⁷ is a tyrosine kinase inhibitor (TKI) that is approved (2020) by the FDA for metastatic or unresectable GIST harboring PDGFRA exon 18 mutations including PDGFRA D842V mutation. Another TKI, imatinib⁶⁸, is approved (2001) for patients diagnosed with chronic eosinophilic leukemia harboring FIP1L1-PDGFRB fusions. Additionally, imatinib is recommended for the treatment of GISTs harboring PDGFRA mutations with the exception of D842V⁶⁹. The TKI, dasatinib, is also recommended for the treatment of GISTs harboring a PDGFRA D842V mutation following disease progression on imatinib, sunitinib, or regorafenib⁶⁹.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E

Relevant Therapy	FDA	NCCN	Clinical Trials*
dabrafenib	☒	☒	×
dabrafenib + trametinib	☒	☒	×
binimetinib + encorafenib	☒	☒	×
cetuximab + encorafenib	☒	☐	×
trametinib	☒	×	×
vemurafenib	☒	☒	☒ (II)
cobimetinib + vemurafenib	☒	☒	☒ (II)
atezolizumab + cobimetinib + vemurafenib	☒	×	×
dabrafenib + pembrolizumab + trametinib	×	☒	×
encorafenib + panitumumab	×	☐	×
bempegaldesleukin, nivolumab	×	×	☒ (III)
dabrafenib, trametinib, ipilimumab, nivolumab	×	×	☒ (III)
encorafenib, binimetinib, pembrolizumab	×	×	☒ (III)
sargramostim, nivolumab, ipilimumab	×	×	☒ (II/III)
abemaciclib + LY3214996	×	×	☒ (II)
atezolizumab, cobimetinib, vemurafenib	×	×	☒ (II)
binimetinib, encorafenib	×	×	☒ (II)
dabrafenib, nivolumab, trametinib	×	×	☒ (II)
dabrafenib, pembrolizumab, trametinib	×	×	☒ (II)
dabrafenib, trametinib	×	×	☒ (II)
dabrafenib, trametinib, spartalizumab	×	×	☒ (II)
encorafenib, binimetinib, nivolumab, ipilimumab	×	×	☒ (II)
IMM-101, nivolumab, ipilimumab	×	×	☒ (II)
immunostimulant, pembrolizumab	×	×	☒ (II)
ipilimumab, binimetinib, nivolumab, encorafenib	×	×	☒ (II)
ipilimumab, nivolumab, encorafenib, binimetinib, dabrafenib, trametinib	×	×	☒ (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
LXH254 , LTT-462, trametinib, ribociclib	×	×	● (II)
nivolumab, ipilimumab, radiation therapy	×	×	● (II)
ulixertinib	×	×	● (II)
vemurafenib, cobimetinib	×	×	● (II)
vemurafenib, cobimetinib, atezolizumab	×	×	● (II)
vemurafenib, cobimetinib, ipilimumab, nivolumab	×	×	● (II)
vemurafenib, cobimetinib, surgical intervention, radiation therapy	×	×	● (II)
vemurafenib, dabrafenib	×	×	● (II)
ASTX029	×	×	● (I/II)
bemcentinib, dabrafenib, pembrolizumab, trametinib	×	×	● (I/II)
dabrafenib, trametinib, antimalarial	×	×	● (I/II)
dabrafenib, trametinib, navitoclax	×	×	● (I/II)
encorafenib, binimetinib, palbociclib	×	×	● (I/II)
HH-2710	×	×	● (I/II)
mirdametinib, lifirafenib	×	×	● (I/II)
vorinostat	×	×	● (I/II)
ABM-1310	×	×	● (I)
AZD-0364	×	×	● (I)
BGB-3245	×	×	● (I)
binimetinib, nivolumab, encorafenib	×	×	● (I)
CART-GD2, vemurafenib	×	×	● (I)
cobimetinib, belvarafenib	×	×	● (I)
DAY-101	×	×	● (I)
E6201	×	×	● (I)
HL-085, vemurafenib	×	×	● (I)
itacitinib, dabrafenib, trametinib	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
JSI-1187, dabrafenib	×	×	● (I)
PF 07284890, binimetinib	×	×	● (I)
RMC-4630	×	×	● (I)
RO-5126766, everolimus	×	×	● (I)
RX208	×	×	● (I)

PDGFRA P895S

Relevant Therapy	FDA	NCCN	Clinical Trials*
imatinib	×	○	×
dasatinib, sunitinib	×	×	● (II)
nilotinib, pazopanib	×	×	● (II)
ponatinib	×	×	● (II)
sunitinib	×	×	● (II)

ARID1A Q766*

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	×	×	● (II)
talazoparib	×	×	● (II)
PLX-2853	×	×	● (I/II)
copanlisib, olaparib, durvalumab	×	×	● (I)
VX-803	×	×	● (I)

IDH2 R172K

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	×	×	● (II)
olaparib	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

IDH2 R172K (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib, ceralasertib	×	×	● (II)
olaparib, durvalumab	×	×	● (II)
talazoparib	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

BRAF V600E

NCT ID	Title	Phase
NCT02224781	DREAMseq (Doublet, Randomized Evaluation in Advanced Melanoma Sequencing) a Phase III Trial	III
NCT04657991	A Phase III, Randomized, Double-blind Study of Encorafenib and Binimetinib Plus Pembrolizumab Versus Placebo Plus Pembrolizumab in Participants With BRAF V600E/K Mutation-Positive Metastatic OR Unresectable Locally Advanced Melanoma	III
NCT03898908	Phase II, Multicentre Clinical Trial to Evaluate the Activity of Encorafenib and Binimetinib Administered Before Local Treatment in Patients With BRAF Mutant Melanoma Metastatic to the Brain.	II
NCT02858921	A Phase II, Randomised, Open Label Study of Neoadjuvant Dabrafenib, Trametinib and / or Pembrolizumab in BRAF V600 Mutant Resectable Stage IIIB/C Melanoma	II
NCT02231775	Neoadjuvant and Adjuvant Dabrafenib and Trametinib in Patients With Clinical Stage III Melanoma (Combi-Neo)	II
NCT04310397	Altering Adjuvant Therapy Based on Pathologic Response to Neoadjuvant Dabrafenib and Trametinib (ALTER-PATH NeoDT)	II
NCT03235245	Combination of Targeted Therapy (Encorafenib and Binimetinib) Followed by Combination of Immunotherapy (Ipilimumab and Nivolumab) vs Immediate Combination of Immunotherapy in Patients With Unresectable or Metastatic Melanoma With BRAF V600 Mutation : an EORTC Randomized Phase II Study (EBIN)	II
NCT04722575	"NEOadjuvant Plus Adjuvant Therapy With Combination or Sequence of Vemurafenib, cobimetinib, and atezolizumab in Patients With High-risk, Surgically Resectable BRAF Mutated and Wild-type Melanoma"	II
NCT02968303	Phase II Study With Combination of Vemurafenib With Cobimetinib in B-Raf V600E/K Mutated Melanoma Patients to Normalize LDH and Optimize Nivolumab and Ipilimumab therapy	II
NCT01989585	Phase I/II Study of Dabrafenib, Trametinib, and Navitoclax in BRAF Mutant Melanoma (Phase I and II) and Other Solid Tumors (Phase I Only)	I/II

Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT04655157	A Multi-Center Phase I/II Open Label Study to Evaluate Safety and Efficacy in Participants With Metastatic BRAF-mutant Melanoma Treated With Encorafenib With and Without Binimetinib in Combination With Nivolumab and Low-dose Ipilimumab. (QUAD 01: Quadruple Therapy in Melanoma)	I/II
NCT02836548	HDAC Inhibitor Vorinostat in Resistant BRAF V600 Mutated Advanced Melanoma	I/II
NCT04190628	A Phase I, First-In-Human, Multicenter, Open-Label Study of ABM-1310, Administered Orally in Adult Patients With Advanced Solid Tumors	I
No NCT ID	Phase I Study of Safety and Immune Effects of an Escalating Dose of Autologous GD2 Chimeric Antigen Receptor-Expressing Peripheral Blood T cells in Patients with Metastatic Melanoma	I
NCT03272464	Phase I Study of INCB039110 in Combination With Dabrafenib and Trametinib in Patients With BRAF-mutant Melanoma and Other Solid Tumors	I
NCT03554083	Neoadjuvant Therapy for Patients With High Risk Stage III Melanoma: A Pilot Clinical Trial	II
NCT03911869	A Phase II, Open-Label, Randomized, Multicenter Trial of Encorafenib + Binimetinib Evaluating a Standard-dose and a High-dose Regimen in Patients With BRAFV600-mutant Melanoma Brain Metastasis	II
NCT04511013	A Randomized Phase II Trial of Encorafenib + Binimetinib + Nivolumab vs Ipilimumab + Nivolumab in BRAF-V600 Mutant Melanoma With Brain Metastases	II
NCT04417621	A Randomized, Open-label, Multi-arm, Two-part, Phase II Study to Assess Efficacy and Safety of Multiple LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic BRAFV600 or NRAS Mutant Melanoma	II
NCT03514901	An Evaluation of the Efficacy Beyond Progression of Vemurafenib Combined With Cobimetinib Associated With Local Treatment Compared to Second-line Treatment in Patients With BRAFV600 Mutation-positive Metastatic Melanoma in Focal Progression With First-line Combined Vemurafenib and Cobimetinib.	II
NCT03754179	A lead-in Phase I Followed by a Phase II Clinical Trial on the Combination of Dabrafenib, Trametinib and the Autophagy Inhibitor Hydroxychloroquine in BRAF/MEK Inhibitor-pretreated Patients With Advanced BRAF V600 Mutant Melanoma	I/II
NCT04720768	A Phase IB Study to Evaluate the Safety, Tolerability and Preliminary Efficacy of the Combination of Encorafenib, Binimetinib and Palbociclib in Patients With BRAF-mutant Metastatic Melanoma (The CELEBRATE Study)	I/II
NCT04249843	A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumors	I
NCT04741997	A Randomized Pilot Trial of Adjuvant Therapy Based on Pathologic Response After Neoadjuvant Encorafenib and Binimetinib in Advanced Melanoma	I
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I

Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT04543188	A Two-Part, Phase Ia/b, Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF 07284890 (ARRY 461) in Participants with Braf V600 Mutant Solid Tumors with and without Brain Involvement	I
NCT03635983	A Phase III, Randomized, Open-label Study of NKTR-214 Combined With Nivolumab Versus Nivolumab in Participants With Previously Untreated Unresectable or Metastatic Melanoma	III
NCT03711188	A Study of the Safety and Efficacy of IMM-101 in Combination With Checkpoint Inhibitor Therapy in Patients With Advanced Melanoma	II
NCT02339571	Randomized Phase II/III Study of Nivolumab Plus Ipilimumab Plus Sargramostim Versus Nivolumab Plus Ipilimumab in Patients With Unresectable Stage III or Stage IV Melanoma	II/III
NCT02910700	A Phase II Study of the TRiPlet Combination of Dabrafenib, Nivolumab, and Trametinib in Patients With Metastatic Melanoma (TRIDeNT)	II
NCT03563729	Efficacy Of Immunotherapy In Melanoma Patients With Brain Metastases Treated With Steroids	II
NCT02872259	A Phase Ib/II Randomised Open Label Study of BGB324 in Combination With Pembrolizumab or Dabrafenib/Trametinib Compared to Pembrolizumab or Dabrafenib/Trametinib Alone, in Patients With Advanced Non-resectable (Stage IIIc) or Metastatic (Stage IV) Melanoma.	I/II
NCT03332589	A Phase I Study Of E6201 For The Treatment Of Central Nervous System Metastases (CNS) From BRAF Or MEK-Mutated Metastatic Melanoma	I
NCT04079166	A Phase II Combination Study of SCIB1 with Pembrolizumab in Patients with Stage III or Stage IV Metastatic Melanoma	II
NCT03340129	A Phase II, Open Label, Single Arm Study of Ipilimumab and Nivolumab With Salvage Radiotherapy in Patients With Melanoma Brain Metastases	II
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
No NCT ID	Phase I study of the safety, tolerability, pharmacokinetics and preliminary efficacy of RX208 in patients with advanced malignant solid tumors	I
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT03155620	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol	II
NCT03220035	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of Vemurafenib in Patients With Tumors Harboring Braf V600 Mutations	II
NCT03781219	A Phase I, Single Arm, Dose Escalation Study to Evaluate Safety, Pharmacokinetics and Preliminary Efficacy of HL-085 Plus Vemurafenib in Patients With BRAF V600 Mutant Advanced Solid Tumor	I
NCT04534283	A Phase II Basket Trial of an ERK1/2 Inhibitor (LY3214996) in Combination With Abemaciclib for Patients Whose Tumors Harbor Pathogenic Alterations in BRAF, RAF1, MEK1/2, ERK1/2, and NF1	II

Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT03239015	Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event	II
NCT02407509	A Phase I Trial of RO5126766 (a Dual RAF/MEK Inhibitor) Exploring Intermittent, Oral Dosing Regimens in Patients With Solid Tumours or Multiple Myeloma, With an Expansion to Explore Intermittent Dosing in Combination With Everolimus	I
NCT04488003	A Two-Part, Phase II, Multi-center Study of the ERK Inhibitor Ulixertinib (BVD-523) for Patients With Advanced Malignancies Harboring MEK or Atypical BRAF Alterations	II
NCT04198818	A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors	I/II
NCT03429803	A Phase I Study of TAK-580 (MLN2480) for Children With Low-Grade Gliomas and Other RAS/RAF/MEK/ERK Pathway Activated Tumors	I
NCT03634982	A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors	I
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I

PDGFRA P895S

NCT ID	Title	Phase
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients With Progressive Locally-advanced or Metastatic Solid Tumors MOST: My own specific treatment	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT02272998	Phase II Study Of Ponatinib For Advanced Cancers With Genomic Alterations In Fibroblastic Growth Factor Receptor (FGFR) And Other Genomic Targets (KIT, Pdgfra, RET FLT3, ABL1)	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II

ARID1A Q766*

NCT ID	Title	Phase
NCT03925350	A Phase II Study of Niraparib in Patients With Advanced Melanoma With Genetic Homologous Recombination (HR) Mutation / Alteration	II
NCT03207347	A Phase II Trial of the PARP Inhibitor, Niraparib, in BAP1 and Other DNA Damage Response (DDR) Pathway Deficient Neoplasms (UF-STO-ETI-001).	II

Clinical Trials Summary (continued)

ARID1A Q766* (continued)

NCT ID	Title	Phase
NCT02286687	Phase II Study of the PARP Inhibitor Talazoparib in Advanced Cancer Patients With Somatic Alterations in BRCA1/2, Mutations/Deletions in Other BRCA Pathway Genes and Germline Mutation in BRCA1/2 (Not Breast or Ovarian Cancer)	II
NCT03297424	A Phase Ib/IIa Dose-escalation Study to Assess Safety, Pharmacokinetics, Pharmacodynamics, and Preliminary Efficacy of PLX2853 in Subjects With Advanced Malignancies	I/II
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and Durvalumab (MEDI4736) in Patients With Advanced Solid Tumors	I
NCT02278250	An Open-Label Study of the Safety, Tolerability, and Pharmacokinetic/Pharmacodynamic Profile of M4344 (Formerly VX-803) as a Single Agent and in Combination With Cytotoxic Chemotherapy in Participants With Advanced Solid Tumors	I

IDH2 R172K

NCT ID	Title	Phase
NCT03212274	A Phase II Study of the PARP Inhibitor Olaparib (AZD2281) in IDH1 and IDH2 Mutant Advanced Solid Tumors	II
NCT03207347	A Phase II Trial of the PARP Inhibitor, Niraparib, in BAP1 and Other DNA Damage Response (DDR) Pathway Deficient Neoplasms (UF-STO-ETI-001).	II
NCT03878095	A Phase II Study of Olaparib and AZD6738 in Isocitrate Dehydrogenase (IDH) Mutant Solid Tumors	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT03991832	A Phase II Study of Olaparib and Durvalumab (MEDI 4736) in Patients With IDH-Mutated Solid Tumors	II

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also

cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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